Satyabrata Das

Software Engineer | Backend, Full-Stack & Machine Learning Systems
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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building production backend services, real-time systems, full-stack applications, and machine learning pipelines. Proficient in Python, C++, TypeScript, SQL, RESTful APIs, AWS, Docker, and CI/CD, with strong foundations in data structures, algorithms, object-oriented design, and concurrency. Track record of owning features from design to deployment and delivering measurable gains in latency, reliability, and load time for products serving 20K+ users. Pursuing an M.S. in Artificial Intelligence Systems at the University of Florida.

KEY SKILLS

- **Languages:** Python, C++, TypeScript, JavaScript, SQL, Swift
- **Backend & APIs:** RESTful APIs, FastAPI, Node.js, Express.js, WebSockets, Real-time Systems, ETL Pipelines
- **Cloud & Databases:** AWS, GCP, PostgreSQL, MySQL, MongoDB, SQL & NoSQL Databases
- **Frontend, Mobile & DevOps:** React, React Native, iOS (Core Data, GCD), Docker, CI/CD, Git, Linux, Grafana
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, OpenCV, MediaPipe, LLM APIs, RAG, LangChain, FAISS

PROFESSIONAL EXPERIENCE

Evy | *Software Engineer, Machine Learning* **May 2026 – Present**
- Owned the first production release of a real-time AI interview monitoring platform end to end, from design to deployment.
- Optimized the in-browser inference and deployment pipeline to sustain 40 FPS on consumer webcams and low-end hardware.
- Strengthened reliability under glare, head pose, and low light via adaptive thresholding across 50 test configurations.
- Developed PyTorch/TensorFlow anomaly detection models over 3 real-time signal streams (gaze, iris, blink).

University of Florida | *Machine Learning Research Assistant* | Gainesville, FL **Mar 2026 – May 2026**
- Created an end-to-end Python data pipeline running MediaPipe 3D pose estimation across 1,000+ clinical patient videos.
- Extracted 3 biomechanical feature sets (joint angles, angular velocity, symmetry) from landmark time-series data.
- Trained regression models on 1,000+ recordings to predict clinician-rated severity scores, automating manual review.

SouthEnd Psychiatry | *IT & Operations Contractor* **Sep 2024 – Aug 2025**
- Automated intake reconciliation with Python scripts, eliminating 30 hours of manual data entry per week.
- Reduced scheduling conflicts by 25% across 5,000+ monthly appointments through automated data validation.
- Maintained EHR systems for 20 clinicians and 10,000+ patient records under strict HIPAA compliance.

ARC Document Solutions | *Software Engineer* **Jul 2022 – Sep 2024**
- Engineered a real-time WebSocket sync layer for 20K+ MAU, reducing latency 25% and support tickets ~30% QoQ.
- Integrated 5 payment gateway APIs (Razorpay, Paytm, Cashfree, Braintree, Elavon), cutting failed transactions 15%.
- Redesigned the data and concurrency layer with GCD multithreading, cutting load time 30% and eliminating UI freezes.

SCI-BI Software Pvt Ltd | *Software Engineer Trainee* **Jul 2021 – Dec 2021**
- Delivered Python and SSIS ETL data pipelines processing 10M+ records daily with 99%+ data accuracy.
- Tuned SQL queries with Prometheus/Grafana monitoring, cutting report time 40% and pipeline failures 25%.

PROJECTS

Albie: Voice-First XR Workspace *(1st Place, Gator Hack V; Best Open Source, MLH)* GitHub
- Built a Meta Quest mixed-reality workspace (Unity 6, Meta XR SDK, MRUK) that anchors to the user's desk, streams live VS Code over WebRTC, and lets an AI agent edit CAD, train models, and draft messages by voice.
- Orchestrated Grok Voice with Claude Agent SDK delegation through a FastAPI/WebSocket hub sharing 60 tools, plus 9 parallel agents in isolated git worktrees with human-approved diffs.
- Automated ML repo onboarding with confidence-scored classification and verified adapter generation, boosting a CIFAR-10 model from 60.6% to 82.0% accuracy by voice alone.

bendwith.us: AI Physical-Therapy Coach Live
- Built an on-device pose-tracking rehab coach (MediaPipe, React, Capacitor) for 5 joints across 3 platforms, with 130+ bilingual ElevenLabs voice cues and 10 Hz session replay for therapists.
- Engineered a FastAPI backend with WebSocket/SSE streaming, storing 100 ms angle data in TimescaleDB (∼11:1 compression) and using Gemini to draft clinician-approved plan changes.

JuggleIQ — *Python, FastAPI, React, YOLO, MediaPipe* GitHub
- Built a GPU-backed FastAPI service for video detection and tracking, with a React dashboard and annotated replay.

EDUCATION

University of Florida Gainesville, FL
Master of Science, Artificial Intelligence Systems Aug 2025 – May 2027 (Expected)
- Coursework: AI Systems, Applied Deep Learning, Machine Learning for AI, Computer Vision & Image Processing

Guru Nanak Institute of Technology Kolkata, India
Bachelor of Technology, Computer Science; GPA: 3.80/4.0 Jul 2018 – Jun 2022
- Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks